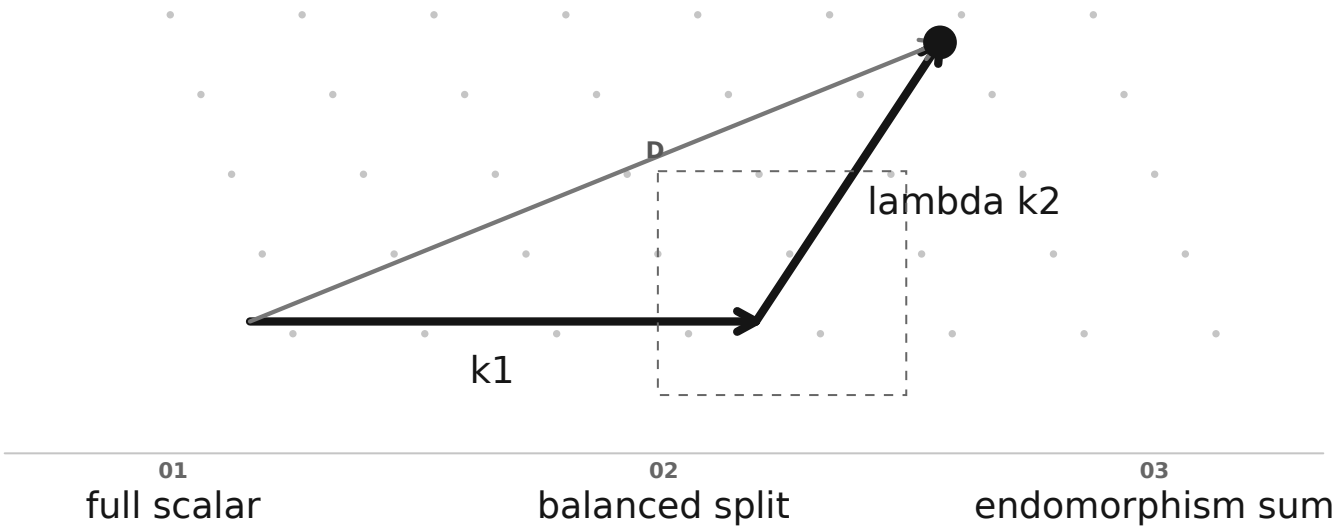
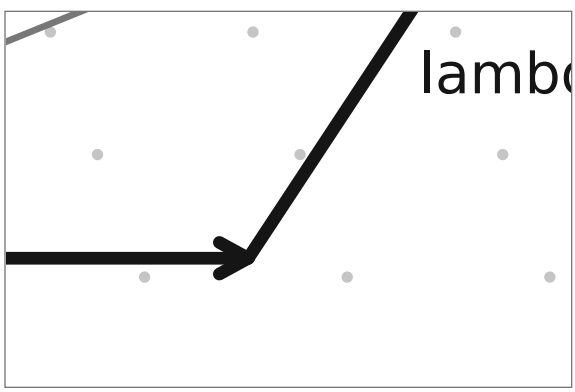
M053

GLV scalar decomposition and endomorphism



D / OPERATION DETAIL



Vector detail / 1.54× main-view scale

HOW THE OPERATION WORKS

The secp256k1 endomorphism permits k congruent to r1+lambda*r2 modulo group order, with a related point map.

READ THE THREE STAGES

01 full scalar
02 balanced split
03 endomorphism sum

WHAT WAS ACTUALLY DONE

EC decomposition code verifies scalar and point identities and can accelerate multiplication. Computing a split of a known scalar does not recover that scalar from a public point.

ORIGINAL FINDING / EXECUTED RECORD

Correct arithmetic acceleration checked on known points; often degenerate for small scalars and not a generic entropy cut.